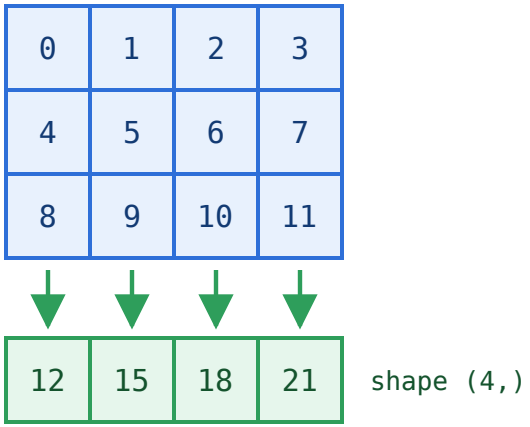
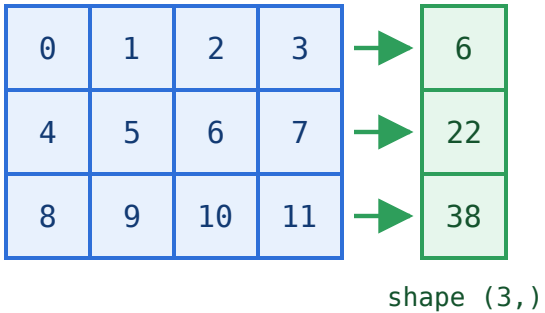
a.sum(axis=0)

collapses axis 0: one result per column



a.sum(axis=1)

collapses axis 1: one result per row



a = np.arange(12).reshape(3, 4); a.sum() with no axis adds everything: 66